

Autrin Hakimi

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PROFESSIONAL PROFILE

Embedded software engineer building real-time automation and control systems for semiconductor manufacturing. I develop Python and C/C++ software coordinating robots, vision sensors, and processing equipment in production workflows. Published researcher (ACM SIGSPATIAL 2025) with experience in distributed computing across 64 NVIDIA A100 GPUs. Strong foundation in POSIX concurrency, socket programming, RTOS concepts, and hardware-software co-design. Seeking to apply embedded systems expertise to high-reliability production environments.

TECHNICAL SKILLS

Embedded & Systems	C/C++, Python, POSIX concurrency (threads, semaphores, mutexes), socket programming, real-time control software, hardware-software integration
Automation & Robotics	Robot workflow orchestration, vision-guided positioning, HMI/GUI diagnostics, UR10e arm, ROS (rospy), OpenCV, MoveIt
Full-Stack & Web	JavaScript, HTML, CSS, React, Python Flask, Tkinter, Node.js, REST APIs, SQL, Android
ML & Data Pipelines	PyTorch, DGL, scikit-learn, distributed training (64-GPU), NetworkX, Pandas, NumPy
DevOps & Tools	Docker, Git, CI/CD pipelines, Linux/Unix, MySQL, Snowflake, Azure, GCP, Jira

EDUCATION

B.S., Computer Science	Aug 2021 - May 2025
Iowa State University GPA: 3.70/4.0 Magna Cum Laude Dean's List	Ames, IA

PROFESSIONAL EXPERIENCE

Systems Engineer, Embedded Software	Jan 2026 - Present
ASM America	Phoenix, AZ
- Build real-time automation software in Python coordinating robots, vision sensors, and semiconductor processing equipment in production manufacturing workflows. - Develop ML-driven inspection models for precision component metrology; build full-stack internal tooling with JavaScript, HTML/CSS frontend and Python Flask backend. - Implement industrial HMI systems supporting diagnostics, operator workflows, and serviceability for embedded equipment platforms. - Design workflow orchestration and error-handling logic for sequential robot operations and vision-guided positioning with sub-millimeter accuracy. - Manage robotics project data using SQL databases; own end-to-end delivery from algorithm design to production deployment. - Partner cross-functionally with mechanical, electrical, and process engineering teams to integrate control software and validate system behavior in production.	
AI Undergraduate Researcher	Aug 2024 - Nov 2025
Iowa State University, SwAPP Lab	Remote
- Co-authored 'HydroGAT' [paper, code] accepted at ACM SIGSPATIAL 2025. - Achieved 15x speedup implementing distributed training across 64 NVIDIA A100 GPUs on the NERSC Perlmutter supercomputer. - Engineered end-to-end data pipelines (PyTorch, DGL, NetworkX, Pandas) for large-scale hydrological datasets; processed geospatial data with xarray and Rasterio. - Automated preprocessing and training workflows via Bash scripting, cutting setup time by 40%.	
Software Development Intern	May 2024 - Aug 2024
BuilderTREND	Omaha, NE
- Designed and shipped a React-based job proposal template; developed full-stack features in JavaScript, TypeScript, and C# (.NET). - Migrated legacy ASP pages to React; collaborated cross-functionally in Kanban/Scrum sprints.	
Data Science Intern	May 2023 - May 2024
Alliant Energy	Remote
- Developed predictive ML models (Random Forest, SVM) analyzing electricity usage data for 995K customers in Snowflake. - Predicted EV ownership for customer segmentation; recognized by Director of Data Analytics for impactful insights.	

SELECTED PROJECTS

Joseph Hoane: Autonomous Robotic Chess System [code]
Built fully autonomous system: real-time piece recognition (OpenCV), move calculation, and UR10e robot arm control (ROS, MoveIt, Python).
xv6-RISC-V Operating System Extensions [code]
Extended xv6 kernel with custom system calls, process scheduling, and memory management in C on RISC-V architecture.
8-Puzzle Solver: AI Search [code]
Implemented BFS, IDS, and A* solvers; achieved 60x speedup via multiprocessing.